

COORDINATE-BLOCK FACTORIZATION UNDER SINGLE-SANDWICH DIVISION AND APPLICATIONS TO PROPERTY $(TT)/T$

ANONYMOUS

ABSTRACT. We prove a coordinate-block factorization theorem for general linear groups over rings satisfying the single-sandwich condition

$$a \neq 0 \implies xay = 1 \text{ for some } x, y.$$

Every element of $\mathrm{GL}_n(R)$ is a product of at most $2n + 2$ factors, each an elementary transvection or an element of one fixed coordinate copy of $\mathrm{GL}_{n-1}(R)$. Thus $\mathrm{GL}_n(R)$ has width at most $2n + 2$ relative to the elementary root subgroups and that fixed coordinate block.

The algebraic factorization is the main input developed here. We combine it with a quantitative root-subgroup estimate in Mimura's relative quasi-cocycle framework. The estimate is proved here by an exhausting finite-word filtration whose boundary terms vanish by an exact telescoping identity. If R is a nontrivial finite-type $\mathbb{F}_2$ -algebra carrying a binary Leavitt family, satisfying the single-sandwich condition, and satisfying $\mathrm{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$, then $E_n(R)$ has property $(TT)/T$ for every $n \geq 2$. For the binary Leavitt algebra, the structural identifications $\mathrm{GL}_n(L) = E_n(L) \cong L^\times$ are known from the work of Khanh–Thanh. Combining them with the rigidity theorem gives the conclusion

$$E_n(L_{\mathbb{F}_2}(1, 2)) \cong L_{\mathbb{F}_2}(1, 2)^\times, \quad n \geq 2,$$

with property $(TT)/T$.

1. INTRODUCTION

All rings and ring homomorphisms in this paper are unital.

Property $(TT)/T$, introduced by Mimura, asks that Hilbert-space quasi-cocycles be bounded after the invariant-vector part of the representation has been removed. It is stronger than Kazhdan's property (T) and is designed to control bounded cohomology and homomorphisms into groups admitting unbounded quasi-cocycles [7]. Ershov and Jaikin-Zapirain proved property (T) for elementary groups of rank at least three over arbitrary finitely generated associative rings [4]. The corresponding quasi-cocycle

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problem is subtler: the bounded error in the cocycle identity accumulates under a long word, so finite generation alone gives no global bound.

The present paper supplies a globalization mechanism for a class of purely infinite rings. Its algebraic input is an explicit matrix factorization. The hypothesis is the following strong two-sided division property:

$$(1) \quad a \neq 0 \implies \exists x, y \in R, \quad xay = 1.$$

We call (1) the single-sandwich property. Ara, Goodearl, and Pardo proved that, for a nondivision simple ring, this is equivalent to pure infiniteness [3, Theorem 1.6].

Theorem A (Coordinate-block factorization). *Let R be a nontrivial ring satisfying (1), let $n \geq 2$, and fix a coordinate $j < n$. Every $A \in \mathrm{GL}_n(R)$ admits a factorization of length at most $2n + 2$ in which every factor is either an elementary transvection or belongs to the coordinate copy*

$$\mathrm{GL}_{n-1}^{(j)}(R) = \{B \in \mathrm{GL}_n(R) : Be_j = e_j, e_j^t B = e_j^t\}.$$

Formal counterpart:

`PropertyTTPaper.coordinateBlock_factorization.`

Theorem A is a relative bounded-width statement rather than bounded elementary generation: the bounded factors come from the elementary root subgroups together with one fixed coordinate block. Fixing the block lets the quasi-cocycle argument use one relative subgroup throughout the factorization. A version in which the coordinate depends on the matrix could also be used after a finite permutation argument, but it would be a weaker width statement with additional bookkeeping. By the characterization just cited, the conclusion applies in particular to every unital purely infinite simple ring.

For the synthesis theorem, a *binary Leavitt family* in R means elements x_0, x_1, y_0, y_1 satisfying

$$y_i x_j = \delta_{ij}, \quad x_0 y_0 + x_1 y_1 = 1.$$

It supplies the matrix self-similarity used to move between ranks. We say that an $\mathbb{F}_2$ -algebra is *finite type* when it is generated by a finite set as an $\mathbb{F}_2$ -algebra, equivalently when it is a quotient of $\mathbb{F}_2\langle X \rangle$ for some finite set X .

Theorem B (Rigidity synthesis). *Let R be a nontrivial finite-type $\mathbb{F}_2$ -algebra. Assume that*

- (1) *R contains a binary Leavitt family;*
- (2) *R satisfies the single-sandwich property (1);*
- (3) *$\mathrm{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$.*

Then $E_n(R)$ has property $(TT)/T$ for every $n \geq 2$.

Formal counterpart:

`PropertyTTPaper.finiteType_elementaryGroup_hasTTmodT.`

The assumptions play separate roles: finite type presents the ring as a quotient of a finite free algebra, the Leavitt family moves between matrix ranks, the single-sandwich property supplies bounded matrix reduction, and the diagonal condition identifies elementary and general linear groups.

There is a useful structural interpretation. Under the first two ring-theoretic assumptions, R is simple, is not a division ring, and is therefore purely infinite simple by Ara–Goodearl–Pardo’s characterization. Their description $V(R) \cong \{0\} \sqcup K_0(R)$ identifies the existence of a binary Leavitt family with the module identity $R \cong R^2$, equivalently $[1_R] = 0$ in $K_0(R)$ [3]. Their isomorphism $R_{\text{ab}}^\times \cong K_1(R)$, together with the GE property, shows that $K_1(R) = 0$ implies the diagonal hypothesis: every unit is a product of commutators, and the elementary identity in Khanh–Thanh’s Lemma 4.1 puts $\text{diag}(u, 1)$ in $E_2(R)$. Thus the two concrete self-similarity and diagonal assumptions are the visible forms of $[1_R] = 0$ and $K_1(R) = 0$ in the purely infinite simple setting.

Theorem C (Binary Leavitt algebra). *The unit group of the binary Leavitt algebra has property $(TT)/T$:*

$$L_{\mathbb{F}_2}(1, 2)^\times \quad \text{has property } (TT)/T.$$

For every $n \geq 2$, matrix self-similarity and the elementary diagonal reduction give

$$E_n(L_{\mathbb{F}_2}(1, 2)) = \text{GL}_n(L_{\mathbb{F}_2}(1, 2)) \cong L_{\mathbb{F}_2}(1, 2)^\times.$$

In particular, every displayed elementary group has property $(TT)/T$.

Formal counterpart:

`PropertyTTPaper.binaryLeavittUnits_hasTTmodT.`

Formal counterpart:

`PropertyTTPaper.binaryLeavitt_elementaryEquivUnits.`

Formal counterpart:

`PropertyTTPaper.binaryLeavitt_elementaryGroup_hasTTmodT.`

The rank-two case is not a low-rank analogue of the universal-lattice argument. The analytic estimate is proved in rank four; the binary Leavitt self-similarity then identifies $E_2(L)$ with $E_4(L)$. This is why a rank-two elementary group can inherit property $(TT)/T$ here.

The algebraic equalities in Theorem C are not new. Khanh and Thanh prove $K_1(L_k(1, 2)) = 0$, perfection of its unit group, and $\text{GL}_n(L_k(1, 2)) = E_n(L_k(1, 2))$ for every $n \geq 2$ [6, Proposition 4.2]. Their complete-prefix-code matrix isomorphisms $M_n(L_k(1, 2)) \cong L_k(1, 2)$ send the elementary group onto the unit group [6, Corollary 4.4]. The contribution of Theorem C is the property $(TT)/T$ conclusion for this already identified unit group.

Relation to previous work. The argument combines several established theories. Mimura introduced property $(TT)/T$, developed the relative-to-global strategy for quasi-cocycles [7]. The quantitative argument below gives an explicit root bound and also controls the invariant-vector summand.

Ershov and Jaikin-Zapirain proved property (T) for elementary groups of rank at least three over finitely generated associative rings [4]. On the ring-theoretic side, Ara, Goodearl, and Pardo obtained the GE consequence for purely infinite simple rings, using the earlier Menal–Moncasi reduction, and identified K_1 with the abelianization of the unit group [8, 3]. Ara, Brustenga, and Cortiñas computed the stable algebraic K -theory of Leavitt path algebras [2].

The pivot in Section 3 belongs to the same bounded-reduction circle as Shalom’s property (T) criterion and the GE reduction used by Ara–Goodearl–Pardo; compare the discussion in Ershov–Jaikin-Zapirain [4]. This is a setting in which stable-range elimination is unavailable: purely infinite simple rings have infinite Bass stable rank [3, Theorem 2.8(2)]. The fixed-coordinate organization is the additional feature used here: the coordinate is chosen before the matrix, exactly one block factor occurs, and the remaining word has length at most $2n + 1$. Thus the full factorization has width $2n + 2$. The rigidity theorem uses this factorization to globalize relative quasi-cocycle bounds. Section 2 gives a quantitative proof of the root estimate, and Section 5 verifies the remaining structural hypotheses for the binary Leavitt algebra.

Each numbered theorem and lemma proved in the formal companion is linked to its public Lean declaration. Appendix A gives a proof-order reading path through the implementation and states the definitions and trust surface used there.

Organization. Section 2 develops the relative root estimate. Section 3 proves the diagonal pivot and the coordinate-block factorization. Section 4 combines the analytic and algebraic inputs to prove Theorem B, and Section 5 verifies its hypotheses for the binary Leavitt algebra. The final section discusses the natural extensions suggested by the proof.

2. QUASI-COCYCLES AND RELATIVE CONTROL

Let G be a group and let $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation on a complex Hilbert space. A map $b : G \rightarrow \mathcal{H}$ is a *quasi-cocycle of defect at most $D \geq 0$* if

$$(2) \quad \|b(gh) - b(g) - \pi(g)b(h)\| \leq D \quad (g, h \in G).$$

An exact cocycle has $D = 0$; it is the translation part of an affine isometric action. A quasi-cocycle permits uniformly bounded failure of the affine action law.

Definition 2.1. A group G has property $(TT)/T$ if, for every unitary representation π with no nonzero invariant vector, every Hilbert-space quasi-cocycle into π is bounded on G . For a subgroup $H \leq G$, the pair (G, H) has relative property $(TT)/T$ if the same hypotheses imply boundedness on H . We write relative (TT) when the conclusion holds without excluding invariant vectors.

We first isolate the quantitative rank-three estimate used in the direct proof. For distinct i, j, k , write $H_{ijk} = \langle X_{ik}, X_{jk} \rangle \leq E_3(A)$, and identify the elements of X with the corresponding free generators of A .

The proof uses the word-length filtration $A_{\leq d} \subseteq A$. Put $V_d = A_{\leq d} \oplus A_{\leq d}$, viewed as the finite elementary abelian subgroup

$$(a, c) \mapsto x_{ik}(a)x_{jk}(c) \quad \text{of } H_{ijk}.$$

For an orthogonal representation σ on a real Hilbert space, let $P_{d,\chi}$ be the spectral projection for $\chi \in \widehat{V}_d$, and put $z_{d,\chi} = P_{d,\chi}z$. The projections are mutually orthogonal and their squared norms sum to $\|z\|^2$. Write a character as (χ_1, χ_2) . For a character χ of $A_{\leq d}$, define

$$\nu(\chi) = \min\{|w| : \chi(w) = -1\},$$

where a word is identified with its monomial in $A_{\leq d}$, and put $\nu(\chi) = d + 1$ when χ is trivial. In particular, $\nu(\chi) = 0$ exactly when $\chi(1) = -1$: degree zero is the one-dimensional constant subspace $\mathbb{F}_2 \cdot 1$. The nontrivial pairs split into

$$\begin{aligned} A & 0 < \nu(\chi_2) < \nu(\chi_1) \\ B & 0 < \nu(\chi_1) = \nu(\chi_2) \\ C & 0 < \nu(\chi_1) < \nu(\chi_2) \\ D & \nu(\chi_1) = 0 \text{ or } \nu(\chi_2) = 0. \end{aligned}$$

Let $m_d^A, m_d^B, m_d^C, m_d^D$ be the corresponding Fourier masses. Let $\partial_{r,d}$ be the mass for which the r -th coordinate character is trivial on $A_{\leq d-1}$ and becomes nontrivial on $A_{\leq d}$. Thus every index below is a genuine character of the finite elementary abelian group V_d .

Lemma 2.2 (Finite-plane energy estimate). *Let $\sigma : E_3(A) \rightarrow \mathcal{O}(\mathcal{H})$ be an orthogonal representation on a complete real Hilbert space. If P is the orthogonal projection onto $\mathcal{H}^{H_{ijk}}$ and $w = z - Pz$, then*

$$\begin{aligned} \|w\|^2 & \leq d_{ik,1}^2 + d_{jk,1}^2 + \frac{3}{2} \sum_{s \in X} 2 \|z\| (d_{ij,s} + d_{ji,s}) \\ (3) \quad & + 2 \|z\| (d_{ij,1} + d_{ji,1}), \end{aligned}$$

where $d_{ab,t} = \|\sigma(x_{ab}(t))z - z\|$.

Formal counterpart:

`PropertyTTPaper.freeCharacteristicTwo_planeEnergyEstimate.`

Proof. Set

$$U_{ab} = 2 \|z\| d_{ab,1}, \quad G_{ab} = \sum_{s \in X} 2 \|z\| d_{ab,s}.$$

Conjugation by $x_{ji}(1)$ shears region A into region B , and conjugation by $x_{ij}(1)$ does the same from C to B . More explicitly, on the plane coordinates these conjugations act by

$$(a, c) \mapsto (a, c + a), \quad (a, c) \mapsto (a + c, c),$$

with the choice of formula determined by the adjacent root. Their dual maps are

$$(\chi_1, \chi_2) \mapsto (\chi_1 \chi_2, \chi_2), \quad (\chi_1, \chi_2) \mapsto (\chi_1, \chi_1 \chi_2).$$

Suppose, for example, that (χ_1, χ_2) lies in A , so $0 < \nu(\chi_2) < \nu(\chi_1)$. At the first degree detected by χ_2 , the character χ_1 is still trivial. Hence $\nu(\chi_1 \chi_2) = \nu(\chi_2)$, while the second character is unchanged, and the image lies in B . Interchanging the coordinates gives the transport from C to B .

For either shear g , spectral covariance is the identity

$$\sigma(g)P_{d,\chi}\sigma(g)^{-1} = P_{d,\chi \circ g^{-1}}.$$

Together with orthogonality and

$$\left| \sum_{\chi \in \Omega} \|z_{d,\chi}\|^2 - \sum_{\chi \in \Omega} \|z'_{d,\chi}\|^2 \right| \leq (\|z\| + \|z'\|) \|z - z'\|$$

for $z' = \sigma(g)z$, these two transports give

$$(4) \quad m_d^A \leq m_d^B + U_{ji}, \quad m_d^C \leq m_d^B + U_{ij}.$$

On region D , one coordinate character detects the unit coefficient. For an involution r , the projection onto its (-1) -eigenspace is $(I - \sigma(r))/2$; hence its squared norm is one quarter of the squared r -displacement. Therefore

$$(5) \quad m_d^D \leq \frac{1}{4}(d_{ik,1}^2 + d_{jk,1}^2).$$

We next record the cross-stage inequalities controlling the equal-valuation region. Write $\text{res}_d : \widehat{V}_{d+1} \rightarrow \widehat{V}_d$ for restriction. For each $\chi \in \widehat{V}_d$, its Fourier component is the orthogonal sum of the components indexed by $\text{res}_d^{-1}(\chi)$. Consequently its squared norm is the sum of their squared norms. A character whose two valuations are not newly exposed in degree $d+1$ remains in the same valuation region after restriction. Therefore, if m_d^S denotes the mass in a union S of valuation regions,

$$(6) \quad m_d^S \leq m_{d+1}^S + \partial_{1,d+1} + \partial_{2,d+1}.$$

The two generator shears give the sharper descents

$$(7) \quad m_{d+2}^{A \cup B} \leq m_{d+1}^{C \cup D} + G_{ji} + \partial_{2,d+2},$$

$$(8) \quad m_{d+2}^{C \cup B} \leq m_{d+1}^{A \cup D} + G_{ij} + \partial_{1,d+2}.$$

We prove the first; the second follows after interchanging the two coordinates. Fix an enumeration of X . For $s \in X$, the relevant shear is the homomorphism

$$\psi_s : V_{d+1} \longrightarrow V_{d+2}, \quad (a, c) \mapsto (a, c + sa).$$

It is induced by conjugation with $x_{ji}(s)$. Notice that its domain and codomain are different: multiplication by s raises degree, so no invariance of a finite truncation is being asserted. Pullback along ψ_s sends a character pair $(\chi_1, \chi_2) \in \widehat{V}_{d+2}$ to

$$\psi_s^*(\chi_1, \chi_2) = (\chi_1 \cdot \partial_s \chi_2, \chi_2)$$

where $(\partial_s \chi_2)(a) = \chi_2(sa)$. If the leading homogeneous part of χ_2 detects a word beginning with s , then

$$\nu(\partial_s \chi_2) = \nu(\chi_2) - 1.$$

No equality is asserted for the other generators. At least one such generator exists: a word detected by the leading homogeneous part has a first letter. Call such a generator *eligible*. Fix an ordering of X and assign the character to its least eligible generator.

Here is the projection bookkeeping for (7). Let $F_s \subseteq \widehat{V}_{d+2}$ be the characters in $A \cup B$ assigned to s . Split

$$F_s = F_s^{\text{int}} \sqcup F_s^{\text{top}},$$

where F_s^{top} consists of characters whose second coordinate becomes visible only in degree $d+2$. The sets F_s^{top} are disjoint and their total mass is at most $\partial_{2,d+2}$. For $\chi \in F_s^{\text{int}}$, the leading part of χ_2 survives restriction to $A_{\leq d+1}$, and

$$\text{res}_{d+1}(\psi_s^* \chi) \in C \cup D.$$

Moreover, the surviving leading part recovers the least eligible letter. Concretely, the second character of $\text{res}_{d+1}(\psi_s^* \chi)$ is the restriction of χ_2 . Below the top boundary its leading homogeneous part is unchanged, so its least eligible generator is still s . If a character belonged to both R_s and R_t , its second coordinate would therefore have least eligible generator both s and t , forcing $s = t$. Thus the receiving sets

$$R_s = \{\text{res}_{d+1}(\psi_s^* \chi) : \chi \in F_s^{\text{int}}\} \subseteq C \cup D$$

are pairwise disjoint.

Let $P_d(\Omega) = \sum_{\chi \in \Omega} P_{d,\chi}$ and put $g_s = x_{ji}(s)$. Spectral covariance says that $\sigma(g_s)P_{d+2}(F_s^{\text{int}})z$ belongs to the range of $P_{d+1}(R_s)$. Apply the variation inequality to $z_s = \sigma(g_s)z$:

$$\left\| P_{d+2}(F_s^{\text{int}})z \right\|^2 \leq \|P_{d+1}(R_s)z\|^2 + 2\|z\| d_{ji,s}.$$

Summing over s , pairwise disjointness gives $\sum_s \|P_{d+1}(R_s)z\|^2 \leq m_{d+1}^{C \cup D}$, so every target component is charged at most once. Adding the top-boundary mass proves (7). The forward shear $(a, c) \mapsto (a + sc, c)$, whose character pullback is $(\chi_1, \chi_2 \cdot \partial_s \chi_1)$, proves (8).

It remains to collect the coefficients. Use (6) to move the two coarse target masses in (7) and (8) to stage $d+2$. Suppressing that common stage and writing $\partial_r = \partial_{r,d+2}$, the resulting inequalities are

$$A + B \leq C + D + G_{ji} + \partial_1 + 2\partial_2,$$

$$C + B \leq A + D + G_{ij} + 2\partial_1 + \partial_2.$$

After addition and cancellation of $A + C$,

$$(9) \quad B \leq D + \frac{1}{2}(G_{ji} + G_{ij}) + \frac{3}{2}(\partial_1 + \partial_2).$$

On the other hand, (4) gives

$$A + B + C + D \leq 3B + D + U_{ji} + U_{ij}.$$

Substitute (9), and then use (5). This gives the completely collected finite-stage inequality

$$(10) \quad m_{d+2}^A + m_{d+2}^B + m_{d+2}^C + m_{d+2}^D \leq d_{ik,1}^2 + d_{jk,1}^2 + \frac{3}{2}(G_{ji} + G_{ij}) + U_{ji} + U_{ij} + \frac{9}{2}(\partial_{1,d+2} + \partial_{2,d+2}).$$

This is the finite-stage estimate whose limit is (3); it also accounts for the coefficient 3/2 there.

The boundary limit follows from an exact operator identity. Let $Q_{r,d}$ be the orthogonal projection onto the vectors fixed by the r -th coordinate copy of $A_{\leq d}$, and put $T_{r,d} = \|Q_{r,d}z\|^2$. Orthogonal refinement gives

$$T_{r,d} = \partial_{r,d+1} + T_{r,d+1}.$$

The nonnegative sequence $T_{r,d}$ decreases, so $\partial_{r,d+1} = T_{r,d} - T_{r,d+1} \rightarrow 0$. Finally, the subgroups V_d increase to H_{ijk} . The orthogonal projections onto their moving subspaces converge strongly to the moving projection for H_{ijk} ; equivalently, the decreasing fixed spaces have intersection $\mathcal{H}^{H_{ijk}}$. Hence the left side of (10) converges to $\|z - Pz\|^2$. Passing to the limit proves (3). $\square$

Lemma 2.3 (Localized plane estimate). *Let $\sigma : E_3(A) \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation and let $z \in \mathcal{H}$. Suppose $B \geq 0$ and the displacement of z is at most B under*

$$x_{ik}(1), x_{jk}(1), x_{ij}(1), x_{ji}(1), \quad x_{ij}(s), x_{ji}(s) \quad (s \in X).$$

Then, for every $g \in H_{ijk}$,

$$\|\sigma(g)z - z\| \leq 2(6|X| + 6)B.$$

Formal counterpart:

PropertyTTPaper.freeCharacteristicTwo_localizedPlaneEstimate.

Proof. Regard the complex Hilbert space as real and the unitary operators as orthogonal operators; norms, fixed spaces, and displacements are unchanged. Let P be the orthogonal projection onto $\mathcal{H}^{H_{ijk}}$, and set $w = z - Pz$. The adjacent root groups X_{ij} and X_{ji} normalize H_{ijk} , while $X_{ik}, X_{jk} \leq H_{ijk}$. Orthogonal projection therefore passes each of the displayed displacement bounds from z to w , with the same bound B . Apply Lemma 2.2 to w . Since $Pw = 0$, it gives

$$\|w\|^2 \leq 2B^2 + (6|X| + 4)\|w\|B.$$

If $B = 0$, this forces $w = 0$. If $B > 0$ and $\|w\| \leq B$, the desired bound on w is immediate. Otherwise $B < \|w\|$, and the last inequality gives $\|w\| \leq (6|X| + 6)B$. Finally, every $g \in H_{ijk}$ fixes Pz , so $\|\sigma(g)z - z\| \leq 2\|w\|$. The formal declaration obtains the same non-strict bound from its strict-bound helper by applying it at $B + \varepsilon/(6|X| + 7)$ and then letting $\varepsilon \downarrow 0$. $\square$

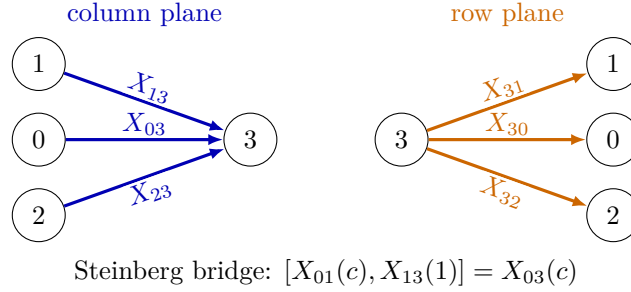


FIGURE 1. The rank-four root geometry. The two rank-three plane estimates control the blue column roots and, through Lemma 2.4, the orange row roots. The displayed Steinberg commutator recovers the target root X_{03} .

Lemma 2.4 (Row-plane symmetry). *Let $^- : A \rightarrow A$ reverse every word and fix the free generators. The inverse-transpose map*

$$\Theta(g) = \bar{g}^{-t}$$

is an automorphism of $E_3(A)$, and

$$\Theta(x_{ij}(a)) = x_{ji}(-\bar{a}) \quad (a \in A).$$

In particular, $\Theta(x_{ij}(t)) = x_{ji}(t)$ for $t \in \{1\} \cup X$. Consequently Lemma 2.3 applies with the two root directions reversed.

Formal counterpart:

`PropertyTTPaper.freeCharacteristicTwo_inverseTranspose_root.`

Formal counterpart:

`PropertyTTPaper.freeCharacteristicTwo_rowEmbedding_controlRoot.`

Formal counterpart:

`PropertyTTPaper.freeCharacteristicTwo_rowEmbedding_elementaryRoot.`

Proof. Word reversal followed by transpose identifies the matrix ring with its opposite ring. On units, inversion identifies the opposite group with the original group, so Θ is an automorphism. Since $E_{ij}^2 = 0$, $x_{ij}(a)^{-1} = x_{ij}(-a)$. In characteristic two the sign vanishes, while word reversal fixes 1 and every $s \in X$, which gives the displayed formula. $\square$

Theorem 2.5 (Root quasi-cocycle estimate). *Let X be finite, put $A = \mathbb{F}_2\langle X \rangle$, and let*

$$G = E_4(A), \quad X_{03} = \{I + aE_{03} : a \in A\}.$$

For every unitary representation π , every $D \geq 0$, and every quasi-cocycle $b : G \rightarrow \mathcal{H}$ of defect at most D , the set $b(X_{03})$ is bounded. Equivalently, the pair (G, X_{03}) has relative property (TT) , with no invariant-vector hypothesis.

Formal counterpart:

`PropertyTTPaper.freeCharacteristicTwo_root_hasRelativeTT.`

Proof. Let N be the last-column subgroup generated by X_{03}, X_{13}, X_{23} . Decompose

$$\mathcal{H} = \mathcal{H}^N \oplus (\mathcal{H}^N)^\perp, \quad b(r) = p + w$$

for $r = I + aE_{03}$. Since the coefficient ring has characteristic two, $r^2 = 1$. First, $\|b(1)\| \leq D$, by applying (2) to $1 \cdot 1$. Apply the same inequality to r^2 and project orthogonally to $\mathcal{H}^N$. Since $r \in N$, it acts as the identity on this subspace. If q denotes the projection of $b(1)$, contractivity of the projection gives

$$\|q - p - p\| \leq D, \quad \|q\| \leq D.$$

Thus $\|2p\| \leq 2D$, and hence

$$(11) \quad \|p\| \leq D.$$

For the moving component, let $Q_X \subseteq E_3(A)$ consist of all elementary roots with coefficient 1 or one of the free generators. Embed it on the coordinates $\{1, 2, 3\}$, and also on the coordinates $\{0, 1, 2\}$ using the transpose–word-reversal symmetry of Lemma 2.4; the geometry is shown in Figure 1. Put

$$M = \sum_{s \in Q_X} (\|b(\iota_{123}(s))\| + \|b(\iota_{012}(s))\|), \quad B = 2M + 2D,$$

$$K = 2(6|X| + 6)B.$$

Only the following members of these two finite embedded sets are used:

$$\begin{array}{l} x_{13}(1), x_{23}(1), x_{12}(t), x_{21}(t), \\ x_{01}(1), x_{02}(1), x_{21}(t), x_{12}(t), \end{array} \quad t \in \{1\} \cup X.$$

Here the first row comes from the 123-embedding and the second from the 012-embedding. Every displayed element commutes with $r = x_{03}(a)$ and normalizes N . Comparing the quasi-cocycle equations for sr and rs , then projecting to the moving subspace, gives for each displayed control element

$$\|\pi(s)w - w\| \leq \|\pi(s)b(r) - b(r)\| \leq 2\|b(s)\| + 2D \leq B.$$

Lemma 2.3 propagates these finitely many inequalities to the complete root planes in the two embeddings. In particular, for every $c \in A$,

$$\|\pi(x_{13}(c))w - w\|, \quad \|\pi(x_{23}(c))w - w\|, \quad \|\pi(x_{01}(c))w - w\| \leq K.$$

The Steinberg relation

$$[I + cE_{01}, I + E_{13}] = I + cE_{03}$$

and two applications of the product displacement inequality give

$$\|\pi([g, h])w - w\| \leq 2\|\pi(g)w - w\| + 2\|\pi(h)w - w\|.$$

With $g = x_{01}(c)$ and $h = x_{13}(1)$, the right side is at most $4K$. Every element of N is a product of one element from each of X_{03}, X_{13}, X_{23} , so its displacement of w is at most $6K$. Since $w \perp \mathcal{H}^N$, this already bounds w . Indeed, the closed convex hull of the N -orbit of w has a unique vector of minimal norm; that vector is N -fixed. The convex hull stays inside $(\mathcal{H}^N)^\perp$, so its minimal vector is zero. On the other hand, the whole orbit, and

hence its closed convex hull, lies in the closed ball of radius $6K$ about w . Consequently $\|w\| \leq 6K$. Together with (11), this gives

$$\|b(I + aE_{03})\| \leq D + 12(6|X| + 6)(2M + 2D),$$

uniformly in $a \in A$. $\square$

3. COORDINATE-BLOCK FACTORIZATION

We now prove the main algebraic input. Write

$$e_{ij}(a) = I + aE_{ij} \quad (i \neq j)$$

for an elementary transvection.

Lemma 3.1 (Three-move diagonal pivot). *Let R satisfy (1), let $A \in \text{GL}_n(R)$, and fix distinct coordinates i, j . At most three elementary row and column operations in total produce a matrix B satisfying*

$$B_{jj} = 1.$$

More precisely, one left factor and two right factors suffice.

Formal counterpart:

PropertyTTPaper.threeMove_diagonalPivot.

Proof. We give the calculation, since it is the source of the uniform constant. Choose two distinct indices i, j . Put $F_0 = I$ if $A_{ii} \neq 0$. If $A_{ii} = 0$, invertibility of A gives k with $A_{ik} \neq 0$; put $F_0 = e_{ki}(1)$. In either case set $A^{(0)} = AF_0$ and $a = A_{ii}^{(0)}$. Then $a \neq 0$, and F_0 is a product of at most one elementary column operation.

Write $b = A_{ij}^{(0)}$, and choose $p, q \in R$ with $paq = 1$. Put

$$\rho = q(1 - pb), \quad F_1 = e_{ij}(\rho), \quad A_1 = A^{(0)}F_1, \quad b_1 = (A_1)_{ij}.$$

Then

$$b_1 = a\rho + b, \quad pb_1 = paq(1 - pb) + pb = 1.$$

Set

$$d_1 = (A_1)_{jj}, \quad w = (1 - d_1)p, \quad E_1 = e_{ji}(w), \quad A_2 = E_1A_1.$$

The (j, j) -entry is now

$$(A_2)_{jj} = d_1 + wb_1 = d_1 + (1 - d_1)pb_1 = 1.$$

Thus $B = A_2 = E_1A(F_0F_1)$ has $B_{jj} = 1$. This uses one left factor and at most two right factors. The calculation applies to arbitrary rings satisfying (1). $\square$

Proof of Theorem A. Choose $i \neq j$. Lemma 3.1 gives elementary products E, F , represented by lists ℓ, r , such that

$$B = EAF, \quad B_{jj} = 1,$$

with

$$|\ell| \leq 1, \quad |r| \leq 2.$$

For every $k \neq j$, right multiply by

$$e_{jk}(-B_{jk}).$$

These transvections have a common source column and distinct target columns, so each previously cleared entry stays zero. For the factor indexed by k , the entry being cleared is

$$(Be_{jk}(-B_{jk}))_{jk} = B_{jk} + B_{jj}(-B_{jk}) = 0.$$

If T is their product and $C = BT$, then

$$C_{jc} = \begin{cases} 1, & c = j, \\ 0, & c \neq j. \end{cases}$$

The list t of right-clearing factors has length exactly $n - 1$.

For every $k \neq j$, left multiply by

$$e_{kj}(-C_{kj}).$$

Again the operations have distinct target rows and preserve the previously cleared entries. If S is their product and $D = SC$, then

$$D_{rc} = \begin{cases} 1, & r = c = j, \\ 0, & r = j, c \neq j, \\ 0, & r \neq j, c = j, \\ C_{rc}, & r, c \neq j. \end{cases}$$

Hence $D \in \text{GL}_{n-1}^{(j)}(R)$. The list s of left-clearing factors also has length exactly $n - 1$.

Solving $D = SEFT$ for A gives

$$A = E^{-1}S^{-1}DT^{-1}F^{-1}.$$

At the word level, inverse products are represented by reversing the factor lists and inverting every transvection. The inverse of $e_{ab}(z)$ is $e_{ab}(-z)$, so all these factors remain elementary. The resulting word has the exact form

$$\ell_{\text{rev}}^{-1} s_{\text{rev}}^{-1} [D] t_{\text{rev}}^{-1} r_{\text{rev}}^{-1},$$

and therefore length at most

$$1 + (n - 1) + 1 + (n - 1) + 2 = 2n + 2.$$

□

Remark 3.2. The fixed coordinate j is chosen before A . Allowing the coordinate block to depend on A gives a weaker width statement. Because there are only finitely many coordinate blocks, that weaker form could still be used after permutation transport; the fixed form keeps one relative subgroup throughout and is the sharper reusable statement.

4. FROM RELATIVE CONTROL TO GLOBAL RIGIDITY

Let R be a finite-type $\mathbb{F}_2$ -algebra. Choose a finite set X and a surjection

$$\mathbb{F}_2\langle X \rangle \rightarrow R.$$

Elementary matrices are functorial in the coefficient ring, so the root estimate of Theorem 2.5 descends to R . Coordinate permutations then give the same estimate for every root subgroup in rank four.

Two larger pieces are used in the assembly: a last-column subgroup and a last-row subgroup. The quasi-cocycle is bounded on both. The stabilized rank-three coordinate group normalizes these pieces. Kazhdan control for the rank-three group converts boundedness on the normalized root system into boundedness on the coordinate group, provided the ambient representation has no invariant vectors. This is the only place where the quotient by the trivial representation enters.

This is the normalized-set step in Mimura's relative-to-global mechanism [7, Theorem 5.1]; the following Kazhdan-pair form records the quantitative estimate used here. For a subset $S \subseteq G$, say that quasi-cocycles are *bounded on S* if every quasi-cocycle into a unitary representation without invariant vectors has bounded restriction to S .

Lemma 4.1 (Kazhdan globalization on a normalized set). *Let G have property (T) , let $S \subseteq G$ generate G , and let $H \leq G$. Suppose*

$$h^{-1}Sh \subseteq S \quad (h \in H)$$

and quasi-cocycles are bounded on S . Then (G, H) has relative property $(TT)/T$.

Formal counterpart:

`PropertyTTPaper.kazhdan_normalizedSet_globalization.`

Proof. Fix a unitary representation π of G without nonzero invariant vectors and a quasi-cocycle b of defect at most D . Choose a Kazhdan pair (K, ε) , and choose once and for all an $S \cup S^{-1}$ -word for each $k \in K$. Let L be the sum of their lengths, so L bounds each chosen length. If $\|b(s)\| \leq C$ for $s \in S$, where C is the bound for this fixed quasi-cocycle, comparison of the quasi-cocycle identities for sh and $h(h^{-1}sh)$ gives

$$\|\pi(s)b(h) - b(h)\| \leq 2C + 2D \quad (s \in S, h \in H).$$

For an inverse letter s^{-1} , apply this estimate to $s \in S$. The identity

$$\|\pi(s^{-1})v - v\| = \|\pi(s)v - v\|$$

holds for every vector v , so with $v = b(h)$ the constant remains $2C + 2D$. No bound on $b(s^{-1})$ is used. The displacement estimate for a product of unitaries therefore gives

$$\|\pi(k)b(h) - b(h)\| \leq L(2C + 2D) \quad (k \in K).$$

Since π has no invariant vectors, the Kazhdan inequality supplies some $k \in K$ with $\varepsilon \|b(h)\| \leq \|\pi(k)b(h) - b(h)\|$. Thus $\|b(h)\| \leq L(2C + 2D)/\varepsilon$, uniformly in $h \in H$. $\square$

Lemma 4.2 (Leavitt rank equivalence). *If R contains a binary Leavitt family, then for all $m, n \geq 2$ there is a group isomorphism*

$$E_m(R) \cong E_n(R).$$

Formal counterpart:

`PropertyTTPaper.leavitt_elementaryRankEquivalence.`

Formal counterpart:

`PropertyTTPaper.elementaryBlockFlattening.`

Proof. For every $q \geq 2$, the complete left-comb binary prefix code

$$\{1^r 0 : 0 \leq r \leq q - 2\} \cup \{1^{q-1}\}$$

has q words. Its Leavitt matrix units give a ring isomorphism $M_q(R) \cong R$. Functoriality in the coefficient ring and block flattening give

$$E_m(R) \cong E_m(M_n(R)) \cong E_{mn}(R) \cong E_n(M_m(R)) \cong E_n(R).$$

The middle identifications are bidirectional on elementary groups. A block transvection decomposes into commuting scalar transvections after flattening. Conversely, a scalar transvection whose indices lie in different outer blocks is already a block transvection. If both indices lie in one outer block, choose a second outer block and use the Steinberg identity

$$e_{(i,k),(i,l)}(a) = [e_{(i,k),(j,k)}(a), e_{(j,k),(i,l)}(1)].$$

Thus block flattening identifies the full elementary subgroups in both directions, and the displayed chain restricts to the claimed group isomorphism. $\square$

Lemma 4.3 (Elementary equals general linear). *Assume that R has a binary Leavitt family, satisfies (1), and satisfies $\text{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$. Then*

$$E_n(R) = \text{GL}_n(R) \quad (n \geq 2).$$

Formal counterpart:

`PropertyTTPaper.elementaryGroup_eq_generalLinear.`

Proof. Apply the pivot and the row and column clearing in the proof of Theorem A with $n = 2$. They reduce an arbitrary element of $\text{GL}_2(R)$, by elementary operations, to the remaining one-by-one coordinate block $\text{diag}(u, 1)$ for some $u \in R^\times$. The diagonal hypothesis therefore gives $\text{GL}_2(R) = E_2(R)$.

The binary Leavitt family gives a coefficient isomorphism $M_n(R) \cong R$ for every $n \geq 2$. Transporting the two ring-theoretic hypotheses across this isomorphism gives the same rank-two equality over $M_n(R)$. The bidirectional block flattening proved in Lemma 4.2 then gives

$$E_2(M_n(R)) = \text{GL}_2(M_n(R)) \implies E_{2n}(R) = \text{GL}_{2n}(R).$$

Reindexing $\text{Fin}(2n)$ by $\text{Fin}(n) \times \text{Fin}(2)$, rather than $\text{Fin}(2) \times \text{Fin}(n)$, gives the same bidirectional flattening in the form

$$E_n(M_2(R)) = \text{GL}_n(M_2(R)).$$

Transport through $M_2(R) \cong R$ now gives $E_n(R) = \text{GL}_n(R)$. $\square$

Proposition 4.4 (Rank-four assembly). *Let X be finite and let $f : \mathbb{F}_2\langle X \rangle \twoheadrightarrow R$ be a surjective ring homomorphism. Suppose that R is nontrivial, contains a binary Leavitt family, satisfies (1), and satisfies $\text{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$. Then $E_4(R)$ has property $(TT)/T$.*

Formal counterpart:

`PropertyTTPaper.finiteFreeQuotient_rankFour_hasTTmodT.`

Proof. Theorem 2.5 descends through f and gives relative $(TT)/T$ for $X_{03}(R)$. Coordinate permutations give it for the six roots

$$X_{03}, X_{13}, X_{23}, X_{30}, X_{31}, X_{32}.$$

The assembly follows the chain

$\begin{aligned} &\text{six root bounds} \implies \text{last row and last column are bounded} \\ &\implies \text{the rank-three core is bounded} \implies \text{coordinate blocks are bounded} \\ &\stackrel{\text{Theorem A}}{\implies} \text{the whole rank-four group is bounded.} \end{aligned}$
--

Here are the details. Put

$$U^+ = X_{03}X_{13}X_{23}, \quad U^- = X_{30}X_{31}X_{32}.$$

The three roots in each product commute, so the quasi-cocycle identity makes b bounded on U^+ and on U^- . The set $S = U^+ \cup U^-$ generates $E_4(R)$: roots meeting the last coordinate lie in S , and every remaining root is a Steinberg commutator of one last-column and one last-row root.

For the Kazhdan input, first apply the Ershov–Jaikin–Zapirain theorem to $E_3(\mathbb{F}_2\langle X \rangle)$ [4]. Property (T) descends along the surjection induced by f , so $E_3(R)$ has property (T) .

The formal proof establishes the same input internally rather than postulating the cited theorem. The declaration `A2MagicHilbert.elementary_exists_rootSet_isKazhdan` provides a Kazhdan constant κ for the union of the six rank-three root subgroups. The plane estimate proves that the finite unit-and-generator control set controls that union with factor $C = 12|X| + 13$. The standard controlled-subset argument then makes this finite set a Kazhdan pair with constant $\kappa/(2C)$. The relevant declaration is

`FreeElementaryPropertyT.controlSet_isKazhdanPair.`

This is the additional step between Lemma 2.3 and a rank-three spectral gap. The resulting pair descends through f . Lemma 4.2 transports property (T) to $E_4(R)$. The stabilized rank-three subgroup H is the upper-left copy of $E_3(R)$, fixing the last basis vector and covector. Conjugation acts on a last-column vector by the upper-left elementary matrix and on a last-row

vector by its inverse. Thus H normalizes both U^+ and U^- , and hence S . Lemma 4.1 now proves that $(E_4(R), H)$ has relative property $(TT)/T$. Notice that the no-invariant-vectors hypothesis is used precisely in the Kazhdan inequality inside that lemma.

By Lemma 4.3, the elementary groups in ranks three and four are the full general linear groups. Under these identifications, the upper-left stabilized subgroup is

$$\{\text{diag}(A, 1) : A \in \text{GL}_3(R)\} = \text{GL}_3^{(3)}(R),$$

the coordinate block fixing the last basis vector and covector. Finally, coordinate permutations carry this relative statement to $\text{GL}_3^{(j)}(R)$ for every $j \in \{0, 1, 2, 3\}$.

Each elementary transvection lies in a coordinate block avoiding its two indices. Hence the union of one fixed coordinate block and all root subgroups is bounded for the quasi-cocycle. Theorem A writes every element of $\text{GL}_4(R)$ as a product of at most 10 elements of that union. Lemma 4.5 below, or directly the corresponding bounded-product lemma, makes the quasi-cocycle bounded on $\text{GL}_4(R)$. Transporting through $E_4(R) = \text{GL}_4(R)$ proves the proposition. $\square$

The passage from a bounded generating subset to the whole group uses the following elementary estimate.

Lemma 4.5 (Products of controlled pieces). *Let b be a quasi-cocycle of defect at most D , and suppose $\|b(g_i)\| \leq C$ for a list $(g_1, \dots, g_m)$. Then*

$$\|b(g_1 \cdots g_m)\| \leq mC + (m+1)D.$$

Formal counterpart:

`PropertyTTPaper.quasiCocycle_list_product_bound.`

Proof. Induct on the list. The empty-list case is $\|b(1)\| \leq D$. For a nonempty list, apply (2) to the head and the product of the tail, use that the representation is isometric, and invoke the inductive bound. The displayed estimate is a slightly loose uniform form that also covers the empty list. $\square$

Proof of Theorem B. Finite type supplies a finite set X and a surjection $\mathbb{F}_2\langle X \rangle \twoheadrightarrow R$, so Proposition 4.4 gives the result in rank four. For arbitrary $n \geq 2$, Lemma 4.2 supplies an explicit group isomorphism $E_n(R) \cong E_4(R)$. Property $(TT)/T$ is invariant under group isomorphism, so the rank-four conclusion transports to rank n . $\square$

5. THE BINARY LEAVITT SPECIALIZATION

Let L be the binary Leavitt algebra [5],

$$L = L_{\mathbb{F}_2}(1, 2) = \mathbb{F}_2\langle x_0, x_1, y_0, y_1 \rangle / (y_i x_j - \delta_{ij}, x_0 y_0 + x_1 y_1 - 1).$$

The defining generators themselves form a binary Leavitt family, and the presentation makes L a finite-type $\mathbb{F}_2$ -algebra.

For a binary word $\alpha = i_1 \cdots i_r$, write $x_\alpha = x_{i_1} \cdots x_{i_r}$ and $y_\alpha = y_{i_r} \cdots y_{i_1}$. Then $y_\alpha x_\beta$ is determined by prefix comparison, and at every depth r ,

$$(12) \quad \sum_{|\delta|=r} x_\delta y_\delta = 1.$$

Lemma 5.1 (Single-sandwich division for $L_k(1, 2)$). *For every field k and every nonzero $z \in L_k(1, 2)$, there are $a, b \in L_k(1, 2)$ such that $azb = 1$.*

Formal counterpart:

`BinaryLeavitt.exists_mul_mul_eq_one.`

Proof. Write z as a finite sum $\sum_i c_i x_{\alpha_i} y_{\beta_i}$, and let $M = \max_i |\beta_i|$. Equation (12) shows that $zx_\delta \neq 0$ for some word δ of length M ; otherwise $z = \sum_{|\delta|=M} zx_\delta y_\delta = 0$. Prefix cancellation expresses zx_δ as a finite linear combination of pure x -words. After collecting equal words, write

$$zx_\delta = \sum_{\gamma \in G} c_\gamma x_\gamma, \quad c_\gamma \neq 0,$$

and choose $\gamma_0 \in G$ of minimal length. Multiplication on the left by y_{γ_0} turns the γ_0 -term into c_{γ_0} ; every other surviving term is a proper x_ε , with $1 \leq |\varepsilon| \leq m$ for $m = \max_{\gamma \in G} |\gamma|$.

Let $w = 0^m 1$. For every such ε , the words w and εw are incomparable. Indeed, εw is longer than w . If w were its prefix, the first m letters of εw would force $\varepsilon = 0^{|\varepsilon|}$, but in position $m+1$ the word w has 1 and εw still has 0. Hence $y_w x_\varepsilon x_w = 0$. It follows that

$$y_w y_{\gamma_0} z x_\delta x_w = c_{\gamma_0}.$$

Thus

$$(c_{\gamma_0}^{-1} y_w y_{\gamma_0}) z (x_\delta x_w) = 1.$$

□

This direct argument proves the single-sandwich form of pure infiniteness used below. Alternatively, the graph criterion of Abrams–Aranda Pino gives pure infiniteness and simplicity, and Ara–Goodearl–Pardo’s characterization gives the same conclusion [1, Theorem 11] [3, Theorem 1.6]. It remains to establish the unstable diagonal condition.

Proposition 5.2 (Elementary matrices over the binary Leavitt algebra). *For every field k , write $L_k = L_k(1, 2)$. For every $u \in L_k^\times$,*

$$\text{diag}(u, 1) \in E_2(L_k).$$

More generally,

$$\text{GL}_n(L_k) = E_n(L_k) \quad (n \geq 2).$$

Formal counterpart:

`KOnePaper.diagUnit_mem_elementary.`

Formal counterpart:

`KOnePaper.elementaryGroup_two_eq_top.`

Formal counterpart:

`BinaryLeavitt.elementaryGroup_eq_top.`

Proof. The Leavitt-path-algebra exact sequence of Ara–Brustenga–Cortiñas applies because k is regular supercoherent. The binary rose has adjacency matrix (2), so the relevant segment is

$$K_1(k) \xrightarrow{-1} K_1(k) \longrightarrow K_1(L_k) \longrightarrow K_0(k) \xrightarrow{-1} K_0(k).$$

This is the specialization of [2, Theorem 7.6]. The two outer maps are automorphisms: under $K_1(k) = k^\times$ the first is inversion, and under $K_0(k) = \mathbb{Z}$ the second is multiplication by -1 . Exactness therefore gives

$$K_1(L_k) = 0.$$

The ring L_k is purely infinite simple. Ara–Goodearl–Pardo show that it is a GE-ring and that the canonical map induces

$$(L_k^\times)_{\text{ab}} \cong K_1(L_k).$$

See [3, Theorem 2.4 and its proof]. Thus $L_k^\times$ is perfect.

It remains only to recall the unstable elementary calculation. For a unit c , put

$$w(c) = e_{12}(c)e_{21}(-c^{-1})e_{12}(c).$$

Direct multiplication gives

$$w(c)w(-1) = \text{diag}(c, c^{-1}) \in E_2(L_k).$$

If $[a, b] = aba^{-1}b^{-1}$, then

$$\text{diag}([a, b], 1) = \text{diag}(a, a^{-1}) \text{diag}(b, b^{-1}) \text{diag}(a^{-1}b^{-1}, ba) \in E_2(L_k).$$

Since every $u \in L_k^\times$ is a product of commutators, this proves $\text{diag}(u, 1) \in E_2(L_k)$. Embedding the calculation in larger matrices shows that every invertible diagonal matrix is elementary. The GE property now gives $\text{GL}_n(L_k) = E_n(L_k)$ for all $n \geq 2$. This is also the argument recorded explicitly by Khanh–Thanh [6, Lemma 4.1 and Proposition 4.2]. $\square$

Remark 5.3 (Constructive formal route). The formal companion also proves Proposition 5.2 by a constructive route, without formalizing algebraic K -theory. Its reduction gives every unit a representative in the degree window $[-1, 1]$, expands that representative relative to complete binary prefix codes into the pencil

$$a_0y_0 + a_1y_1 + c + b_0x_0 + b_1x_1,$$

and applies the scalar-matrix dichotomy to the stacked pair $\begin{pmatrix} B_0 \\ B_1 \end{pmatrix}$. If the row and column codes have cardinalities r and c , then $B_0, B_1 \in M_{r \times c}(k)$, so this stack is the linear map $k^c \rightarrow k^{2r}$. It either has a nonzero kernel, or it is injective and hence has a left inverse. A kernel vector produces a B -free column; splitting its code word increases the column-code cardinality by one. A left inverse instead forces all entries of the inverse pencil into

strictly negative degree. The two branches terminate respectively at the deep-column and inverse deep-row nonpositive-window lemmas. The decreasing counter is $2^{r+1} - |Q|$, with the row code fixed at cardinality 2^r . This is the exact route through `exists_narrow_representative`, `exists_pencil_form`, `pencil_free_exit`, `pencil_full_exit`, and `pencil_unit_mem_pow` listed in the formal-companion reading guide.

Proof of Theorem C. The binary Leavitt algebra is a nontrivial finite-type $\mathbb{F}_2$ -algebra, its canonical generators form a binary Leavitt family, Lemma 5.1 supplies (1), and Proposition 5.2 supplies the elementary diagonal condition. Theorem B gives property $(TT)/T$ for $E_n(L)$ in every rank $n \geq 2$, while Lemma 4.3 identifies it with $GL_n(L)$.

For $n = 1$, take the one-word code consisting of the empty word. For each $n \geq 2$, the left-comb prefix code

$$C_n = \{1^r 0 : 0 \leq r \leq n - 2\} \cup \{1^{n-1}\}$$

is a complete binary prefix code with n words. Its Leavitt matrix units give a ring isomorphism $M_n(L) \cong L$, hence a group isomorphism $GL_n(L) \cong L^\times$. For $n \geq 2$ this yields $E_n(L) \cong L^\times$. Transporting the rank-two conclusion across this isomorphism gives property $(TT)/T$ for $L^\times$. $\square$

6. FURTHER DIRECTIONS

The characteristic-two hypothesis enters most visibly in (11), where every root element has order two. Over a field of characteristic p , the corresponding root has order p , and iterating the quasi-cocycle identity gives

$$b(1) \approx b(r) + \pi(r)b(r) + \cdots + \pi(r)^{p-1}b(r).$$

On the root-fixed space this is p times the fixed projection of $b(r)$, with error at most $(p - 1)D$. Thus the fixed-space argument has a plausible finite-characteristic analogue. Extending the theorem requires uniform replacements for the characteristic-two rank-three control and the sign-free coordinate symmetries.

The algebraic factorization theorem suggests a separate question. For which natural classes of simple or purely infinite rings does the single-sandwich condition hold, and when can the coordinate block in Theorem A itself be controlled by elementary subgroups? A satisfactory answer would extend the present synthesis from the Leavitt setting to a broader rigidity criterion. The binary-family hypothesis is restrictive: it gives an isomorphism of right modules $R \cong R^2$, hence $[R] = 2[R]$ in $K_0(R)$. For $L_k(1, d)$, the class of the unit generates $K_0 \cong \mathbb{Z}/(d - 1)$ [2], so no binary Leavitt family can exist when $d > 2$. Theorem B therefore does not extend to the higher-arity Leavitt algebras without a different self-similarity input.

APPENDIX A. GUIDE TO THE FORMAL COMPANION

The accompanying Lean formalization is included in the supplementary material. Except where indicated, declarations are in the `PropertyTTPaper`

namespace. Each numbered result carries a visible link to its public declaration. The table below lists the main implementation declarations in the order used by the mathematical argument. It is intended as a reading path for users who want to inspect a calculation or reuse a proof component.

Argument	Principal Lean implementation declarations, in proof order
Finite-plane filtration and Lemma 2.2	<code>FreeRootCharacterValuation.characterValuation_eq_zero_iff;</code> <code>planeCharacterRegion_oppositeUnitConjugatedRestriction_eq_B_of_valid;</code> <code>planeCharacterRegion_forwardUnitConjugatedRestriction_eq_B_of_valid;</code> <code>sum_norm_planeRegionSignSet_A_sq_le_B_add_unit_error;</code> <code>sum_norm_planeRegionSignSet_C_sq_le_B_add_unit_error;</code> <code>sum_norm_planeRegionSignSet_D_sq_le_unit_displacements;</code> <code>sum_norm_planeRegionUnion_sq_le_succ_add_topBoundaries;</code> <code>planeABOppositeInteriorImageSignSet_pairwise_disjoint;</code> <code>sum_norm_planeABInteriorLeadingSignSet_sq_le_target_add_error;</code> <code>planeCBForwardInteriorImageSignSet_pairwise_disjoint;</code> <code>sum_norm_planeCBInteriorLeadingSignSet_sq_le_target_add_error;</code> <code>sum_norm_planeABRegion_sq_le_target_add_error_add_boundary;</code> <code>sum_norm_planeCBRegion_sq_le_target_add_error_add_boundary;</code> <code>planeRegionMass_B_le_D_add_generatorErrors_add_boundaries;</code> <code>sum_planeRegionMass_nonzero_le_explicit_errors;</code> <code>tendsto_planeFirstTopBoundaryMass_succ_zero;</code> <code>tendsto_planeSecondTopBoundaryMass_succ_zero;</code> <code>tendsto_norm_planeMovingPart_sq</code>
Lemma 2.3	<code>norm_comp_columnPlane_sub_le_of_root_bounds</code>
Row-plane transport and Theorem 2.5	<code>FreeElementaryTranspose.elementaryTransposeEquiv_elementaryRoot;</code> <code>RankFourRootGeometry.embedRow012_elementaryRoot;</code> <code>RankFourRelativeTT.norm_root03_movingProjection_le;</code> <code>RankFourRelativeTT.norm_root03_fixedProjection_le;</code> <code>RankFourRelativeTT.hasRelativeTT_X03</code>
Theorem A	<code>exists_three_move_diagonal_pivot;</code> <code>rightClearFactors_clear_row;</code> <code>leftClearFactors_make_coordinateBlock;</code> <code>boundedProduct_coordinateBlock0rRoot</code>
Rank transport and rank-four assembly	<code>A2MagicHilbert.elementary_exists_rootSet_isKazhdan;</code> <code>FreeElementaryPropertyT.controlSet_controls_rootSet;</code> <code>FreeElementaryPropertyT.controlSet_isKazhdanPair;</code> <code>FreeElementaryPropertyT.freeElementary_hasKazhdanPropertyT;</code> <code>LeavittFamily.rankSuccEquiv; elementaryBlockEquiv;</code> <code>FiniteTypeLeavittTT.elementaryGroup_eq_top;</code> <code>RankFourGlobalization.hasRelativeTTmodT_coreRange_of_roots;</code> <code>FiniteTypeLeavittTT.rankFour_hasTTmodT;</code> <code>FiniteTypeLeavittTT.elementaryGroup_hasTTmodT</code>
Proposition 5.2	<code>BinaryLeavitt.exists_narrow_representative;</code> <code>BinaryLeavitt.exists_pencil_form;</code> <code>BinaryLeavitt.pencil_free_exit; BinaryLeavitt.pencil_full_exit;</code> <code>BinaryLeavitt.pencil_unit_mem_pow;</code> <code>BinaryLeavitt.narrowReduction_holds;</code> <code>BinaryLeavitt.stableUnits_eq_top_holds;</code> <code>BinaryLeavitt.glTwo_eq_elementary_holds;</code> <code>BinaryLeavitt.elementaryGroup_eq_top</code>
Lemma 5.1	<code>LeavittFamily.eq_sum_mul_wordS_mul_wordT;</code> <code>BinaryLeavitt.killWord_not_prefix;</code> <code>BinaryLeavitt.exists_mul_mul_eq_one</code>

Trust surface and definitions. The executable audit `scripts/Audit.lean` traverses the kernel dependency closure of every declaration in the `GroupApproximation` namespace. It rejects `sorryAx`, compiler-trust axioms, and project axioms; the current audit succeeds with only `propext`, `Classical.choice`, and `Quot.sound`, the three classical axioms used by Mathlib. There are no `sorry` terms in the development.

The definitions used by the linked statements are in `Basic.lean` in the `PropertyTT` directory. The public theorems are universe-polymorphic: they quantify over complete complex Hilbert spaces at an arbitrary universe level and over unitary representations; `IsQuasiCocycle` includes $D \geq 0$ and (2); `HasTTmodT` excludes nonzero invariant vectors exactly as in Definition 2.1. The rank-three Kazhdan input is proved from an explicit finite Kazhdan pair; its implementation entry point appears in the rank-transport row of the table. The single-sandwich and unstable diagonal inputs for $L_k(1, 2)$ are theorems, not hypotheses, in `LeavittSimplicity.lean` and `RefineLoopDischarge.lean`. The latter uses the constructive pencil reduction of Remark 5.3; the formal development does not encode the cited K -theory exact sequence or GE theorem as axioms.

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